A Micro Project report on

"CREATE GEOGRAPHIC DATA VISUALIZATION USING TABLEAU"

Submitted to CMR Institute of Technology, Kandlakoya , Hyderabad In partial fulfilment of the requirements for the award of the Laboratory of

DATA EXPLORATION AND VISUALISATION

Of

II-B.Tech. I-Semester

in

Computer Science and Engineering (CSE)

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(Approved by AICTE, Affiliated to JNTU, Kukatpally, Hyderabad)

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CERTIFICATE

This is to certify that a Micro Project entitled with ""CREATE GEOGRAPHIC DATA VISUALIZATION USING TABLEAU" is being

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In partial fulfillment of the requirement for the completion of the "DATA EXPLORATION AND VISUALISATION" of II-B.Tech, I-Semester, this report is a record of bona fide work carried out under guidance and supervision.

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ABSTRACT

Geographic visualization places measurements in their spatial context, allowing differences between locations to be seen and compared. This micro project examines the use of map visuals for representing state and union territory literacy rates in India. The supplied CSV contains 35 records for Census Year 2011, with 28 states and 7 union territories, and reports literacy as a percentage.

The project uses Tableau to explore geographical data through a state bar chart, point map, dashboard, and data-source view. These visualizations help compare literacy rates across Indian states and union territories. Three charts generated from the CSV further compare literacy rates by location, summarize state and union territory groups, and show the overall distribution.

Across all 35 units, the mean literacy rate is 77.85% and the median is 76.9%. Kerala has the highest recorded value (94.0%), while Bihar has the lowest (61.8%), a range of 32.2 percentage points. The maps in the supplied screenshots show geographic marks, but the screenshots do not expose readable values for each mark and indicate two unknown locations. The tabular and chart analysis supplies the precise comparisons; no causal explanation is inferred from the descriptive differences.

INTRODUCTION

1. Geographic data visualization

Geographic data visualization represents observations according to a place, area, or coordinate. It joins a value to a spatial reference so that a reader can assess where a pattern occurs. A geographic field may be a named administrative unit, an address, or a latitude and longitude pair. The accuracy of the resulting view depends on the quality and specificity of that reference.

A map can help identify spatial concentration, regional variation, and gaps in coverage. It also provides context that a table alone does not show. However, a map should be paired with labels, legends, and a scale or tooltip so that its encoding can be interpreted. Geographic similarity does not by itself explain why measurements differ.

2. Map types and their uses

Visual type	Typical encoding	Use in this project
Bubble / point map	A marker is placed at a geocoded location; size or color can show a measure.	Show the distribution of state and UT records across India.
Filled / choropleth map	A region boundary is shaded according to a numeric value.	Compare literacy rates by state/UT, if region boundaries and correct matches are available.
Bar chart	Length encodes numeric value on a common scale.	Rank states and UTs precisely where map labels are difficult to read.

Microsoft Power BI provides several map options, including bubble and filled map visuals. A point map is useful for locations represented by markers; a filled map emphasizes values attached to defined regions. This report discusses these concepts, while accurately labeling the supplied application screenshots as Tableau evidence.

3. Why map literacy rates

Literacy is a percentage, so it can be compared across administrative units without relying on population totals. Placing these percentages on a map can reveal whether higher and lower values appear in particular parts of the country. The supplied records contain names for states and union territories but no coordinates or polygon boundaries. A mapping product may geocode those names, yet the geocoding result must be checked, especially for historical or abbreviated names.

A bubble map and a filled map answer related but distinct questions. A bubble map plots a marker at a location and can show a value through its size or color. A filled map colors an area and is easier to compare when boundary data is available. For a single measure such as literacy rate, the legend should use a continuous scale with clear endpoints and units. An accompanying ranked chart helps readers compare close values accurately.

4. Scope and interpretation

The analysis is descriptive and limited to the 2011 figures in the supplied file. It does not combine the literacy rate with population, gender, age, or other explanatory variables. Therefore, the report compares recorded rates among locations but does not claim to measure causes, change over time, or the number of literate people.

The screenshots provided for this report visibly use Tableau, including its worksheet tabs, Data pane, Show Me panel, and dashboard controls. They document a map and bar-chart exploration of the supplied data. The report distinguishes those screenshots from the Power BI map concepts in its title and does not claim that the images came from Power BI.

DATASET DESCRIPTION

Dataset overview

The supplied comma-separated file contains 35 rows and 6 columns. It represents India for Census Year 2011 and includes 28 states and 7 union territories. Each row has one administrative unit and one literacy-rate percentage. There are no missing cells and no fully duplicated rows in the file.

Dataset property	Observed value
Records / fields	35 records / 6 fields
Administrative units	28 States; 7 Union Territories
Country and year	India; 2011 for every record
Missing values	0 across all six columns
Exact duplicate rows	0
Geographic detail	Names of state/UT only; no coordinates or boundary geometry
Measurement	Literacy rate, in percent

Data structure

The field names and descriptions below reflect the supplied CSV. The Source field contains the same cited source for all records. Literacy_Rate_Percent is numeric and suitable for ranking or color/size encoding. State_or_UT is categorical and is the principal geographic identifier.

Column name	Data type	Description
State_or_UT	Text	Name of the state or union territory.
Administrative_Unit	Text	Administrative classification: State or Union Territory.
Country	Text	Country name; India for all records.
Literacy_Rate_Percent	Decimal	Literacy rate expressed as a percentage.
Census_Year	Integer	Census year associated with the reported rate; 2011 throughout.
Source	Text	Citation and URL supplied with the dataset.

Table 1: Fields in the supplied literacy dataset

Data quality and mapping suitability

The file has complete records and no repeated rows. The percentage values range from 61.8 to 94.0. State and UT labels are usable as categorical geographic names, but the file does not provide a separate country code, latitude/longitude, or boundary key. Several labels reflect the 2011 administrative structure and abbreviations, including "A & N Islands," "D & N Haveli," and "Daman & Diu." These can be ambiguous to an automated geocoder and should be checked against the intended historical boundaries.

The supplied screenshots show 33 marks and a message indicating two unknown locations. This visible count is consistent with incomplete geocoding in the shown map view, though the screenshots alone do not identify which two names failed. The CSV includes 35 records; consequently, a map with 33 marks should not be treated as a complete representation. The bar chart and generated ranking chart retain all 35 records.

A filled map would require valid state/UT boundaries corresponding to the 2011 units. Without geometry or a verified boundary matching layer, this report does not generate a filled map image. The three generated charts use the CSV values directly and make no geographic-coordinate assumptions.

OBJECTIVES

The objectives of this micro project are:

- To understand how geographic visualization relates measurements to administrative locations.
- To inspect the supplied CSV structure, data types, coverage, and completeness.
- To identify the state/UT field as the geographic dimension and literacy percentage as the numeric measure.
- To examine point-map and filled-map concepts available in Power BI and understand the role of geographic categorization.
- To interpret the supplied map and dashboard screenshots without overstating what the images establish.
- To compare literacy rates across all 35 states and union territories using readable charts and summary statistics.
- To report the highest and lowest values and describe the limits of mapping records with names but no coordinates or boundaries.
- To present the observations in a clear academic report with justified explanatory paragraphs.

PROJECT ANALYSIS AND IMPLEMENTATION

Dataset import and inspection

The CSV is the primary analysis file. Its header row defines six fields, and its 35 data rows were inspected for column types, missing values, and exact duplicates. Literacy_Rate_Percent was treated as a numeric measure; State_or_UT and Administrative_Unit were treated as categories. Census_Year and Country were checked for consistency.

The supplied data-source screenshot shows a text-file connection and a preview table with field names. The interface shown is Tableau. It supports the observation that the file was loaded into a visualization environment, but it is not evidence of a Power BI import. The report's calculated summaries were independently derived from the supplied CSV.

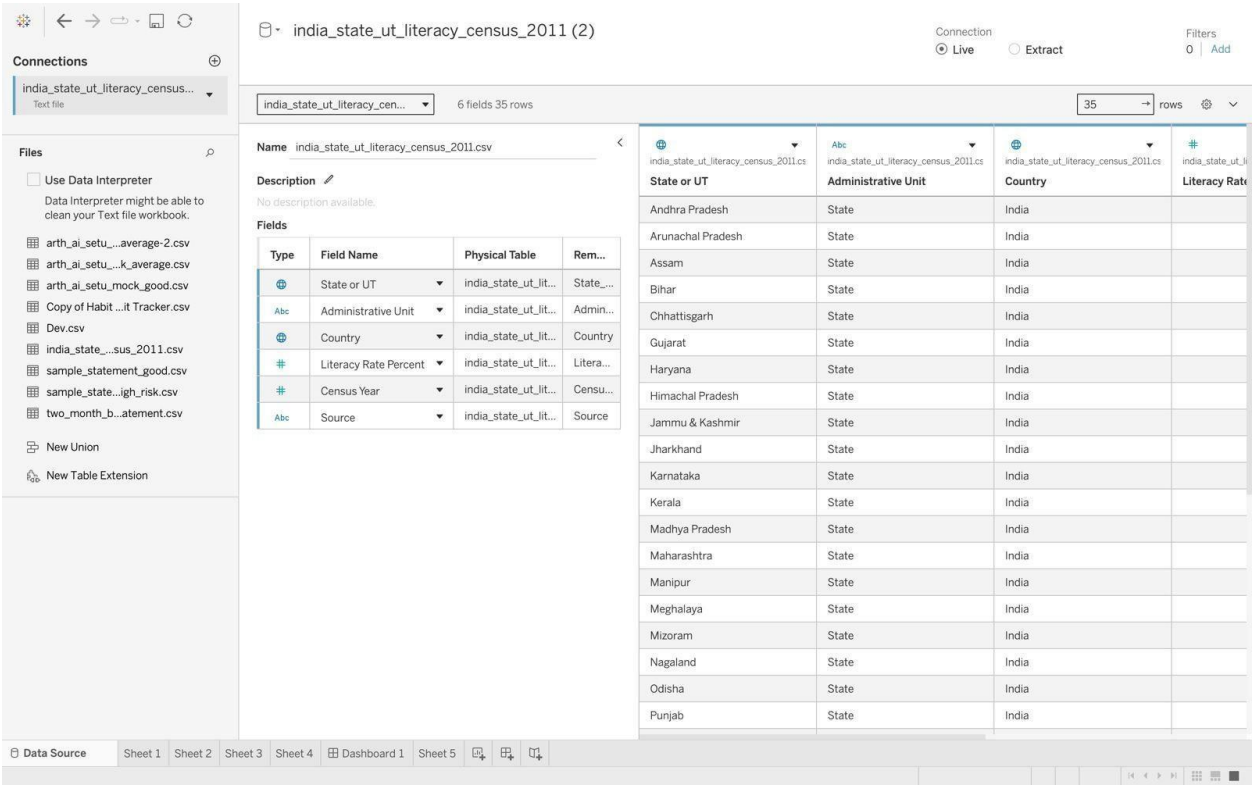


Figure 1: CSV field list and preview in the supplied Tableau data-source screenshot

Observation: The preview shows state/UT names, administrative-unit categories, country, and the literacy-rate field. The CSV contains 35 records, while the visible preview is only a portion of the rows. No latitude or longitude columns appear in the supplied file.

Geographic field selection and point map

For a state-level map, State_or_UT is the location field and Literacy_Rate_Percent is the measure of interest. In Power BI, a geographic data category and a country context help distinguish similarly named places. The marker or bubble layer can encode the measure through color or size, while tooltips can expose the exact unit name and rate. Because all records concern India, the Country field can provide additional disambiguation.

The supplied map screenshot contains blue circular marks on a basemap of the Indian subcontinent. The map has 33 marks and a visible “2 unknown” indicator. The screenshot does not show value labels on the markers, so it supports a geographic distribution reading but not exact comparisons between individual literacy rates.

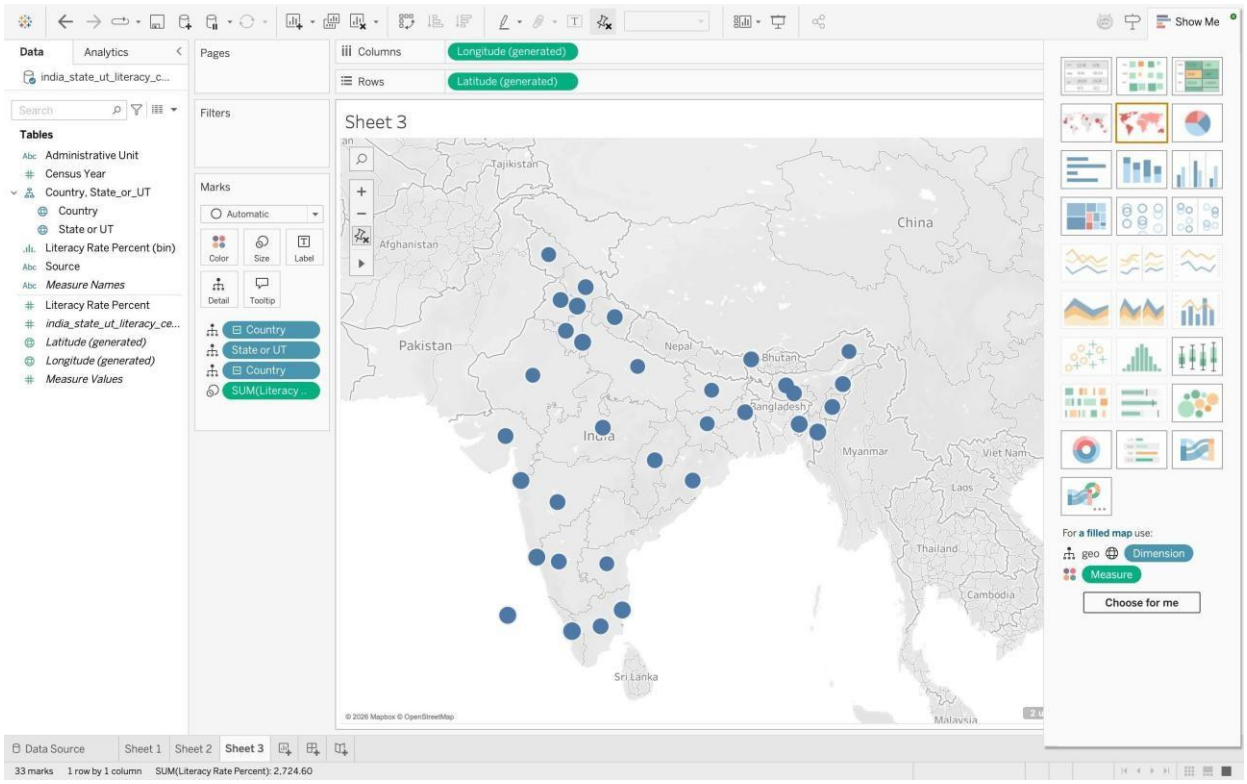


Figure 2: Point map of the state and UT records in the supplied Tableau worksheet screenshot

Observation: The map places marks across the country, including a dense cluster in the northeast where several units are geographically close. Two records are reported as unknown by the visual. Since the image does not identify the unmatched rows, those locations should be diagnosed using a location table and checked names before interpreting spatial coverage.

Comparing geographic values with a bar chart

A bar chart complements a map by placing every value on a shared axis. The supplied worksheet screenshot displays state/UT names on the horizontal axis and a numeric scale from approximately 71 to 96. It shows substantial variation among locations, but labels are rotated and the screenshot does not make every precise value easy to read. The CSV-based ranking chart resolves this limitation by displaying all 35 values with direct labels.

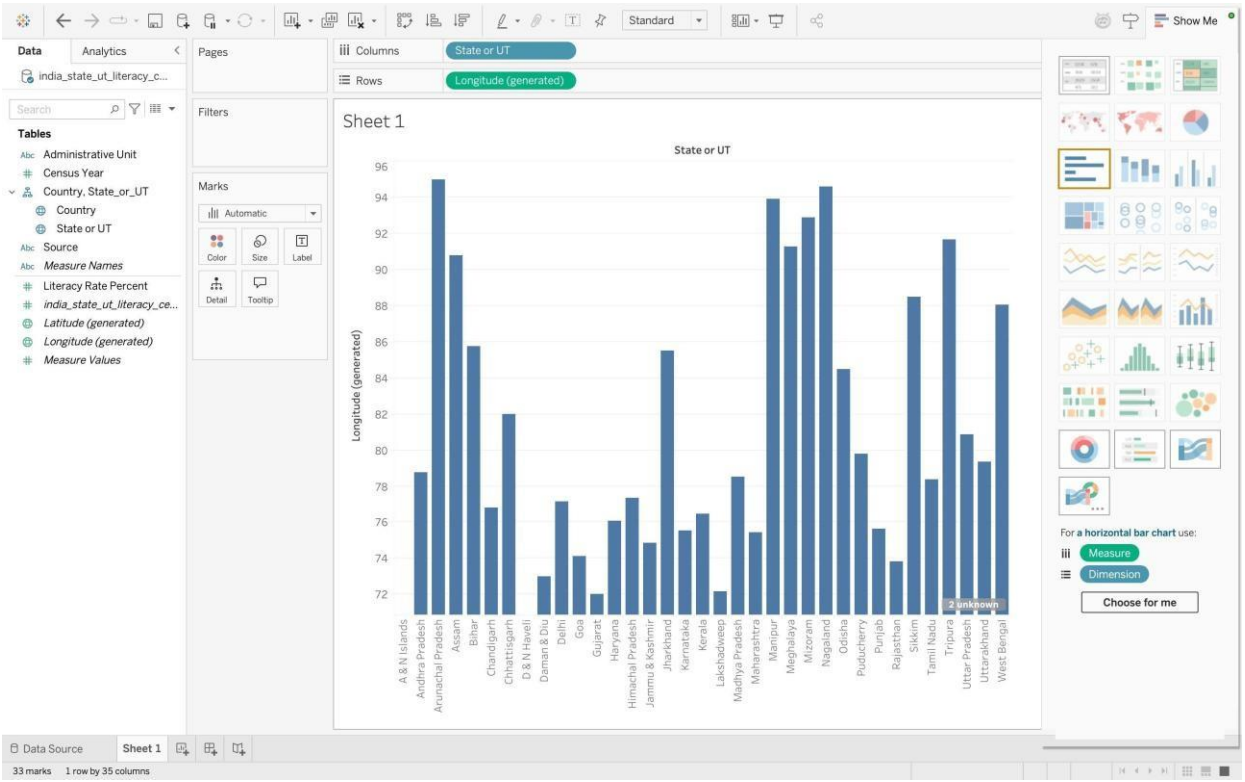


Figure 3: State/UT literacy-rate bar chart in the supplied Tableau worksheet screenshot

Observation: The visible chart shows a wide spread, with the tallest bars near the mid-90s and the shortest near 62%. The screenshot also displays an “unknown” count associated with geographic fields; the bars themselves remain a useful comparison of the rate values. Exact top and bottom values are calculated from the CSV in Section 6.

Using map and chart together

The dashboard screenshot places a point map beside a state/UT bar chart and a legend. The paired views are useful because the map supplies spatial context while the chart supports value comparison. The dashboard shown is Tableau; no Power BI dashboard capture was supplied. The same design can be implemented in Power BI with a map visual, a bar chart, and shared filters or cross-highlighting.

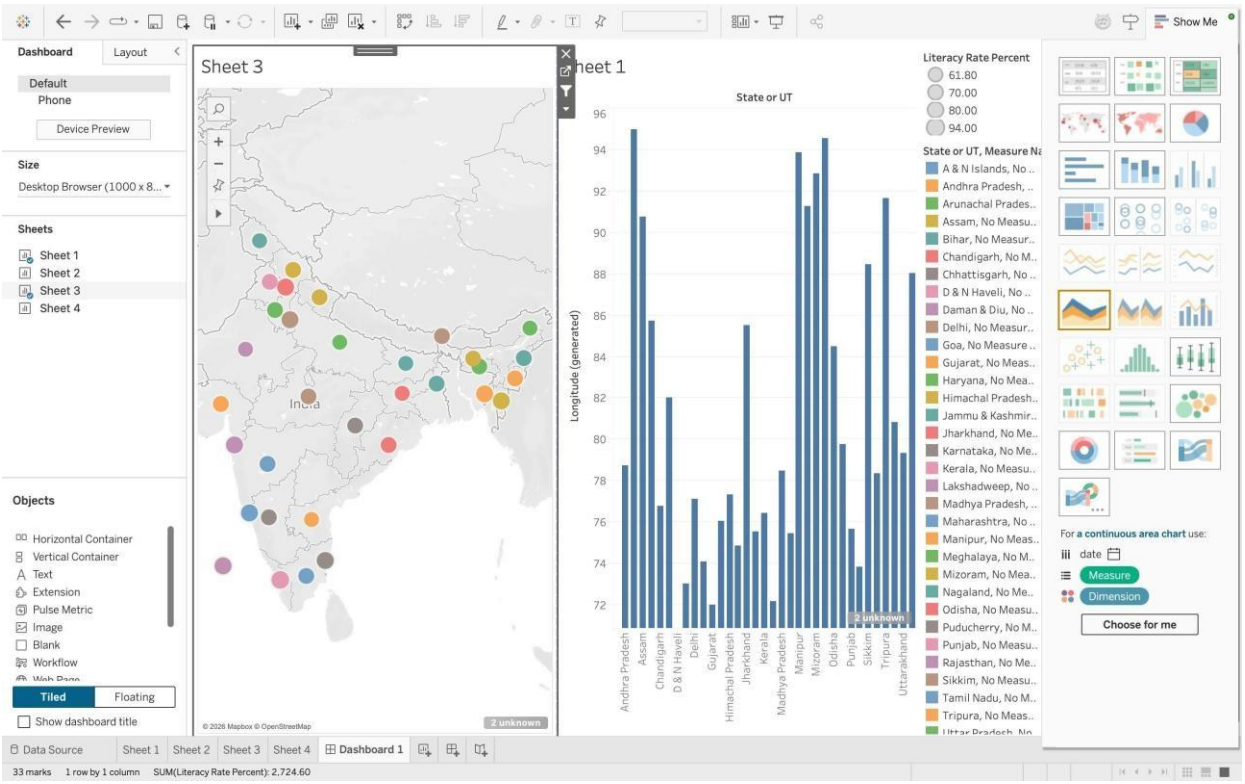


Figure 4: Map and bar chart arranged in the supplied Tableau dashboard screenshot

Observation: The dashboard combines a map with a long state/UT bar chart and a legend. The map is constrained to a narrow pane, and the legend consumes substantial space. A practical revision would widen the map, reduce or replace the long legend, and use a sorted horizontal ranking chart with visible rate labels. The screenshot shows a dashboard composition, not a Power BI interface.

Additional chart generated from the CSV

The report's ranking chart sorts all 35 records by literacy rate and uses a distinct color for states and union territories. This chart is generated from the supplied CSV and is not a screenshot of a software dashboard. It helps verify comparisons that are difficult to read in the supplied worksheet image.

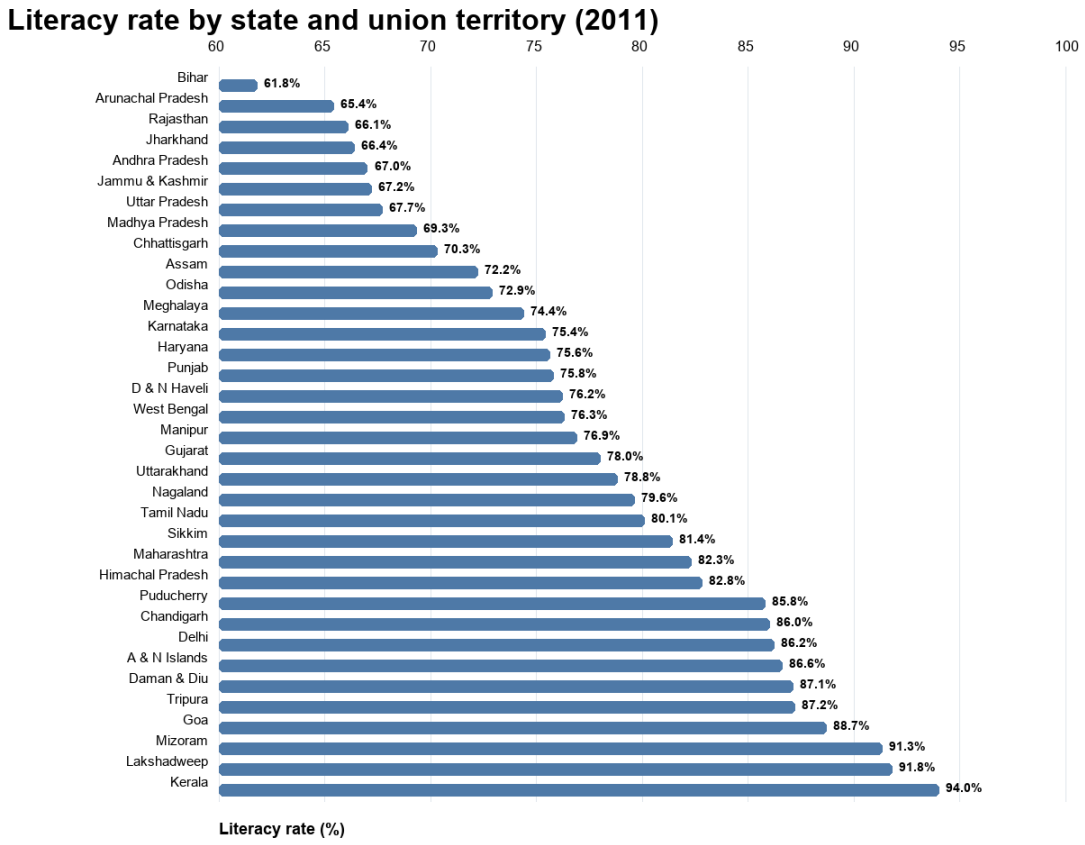


Figure 5: Literacy-rate ranking for all 35 states and union territories (generated from CSV)

RESULTS AND OBSERVATIONS

Summary statistics

The mean literacy rate across all 35 administrative units is 77.85%, while the median is 76.9%. The minimum is 61.8% in Bihar and the maximum is 94.0% in Kerala. The difference between these endpoints is 32.2 percentage points. These are unweighted summaries across units; they are not population-weighted national literacy estimates.

Statistic	Value
Number of records	35
Mean	77.85%
Median	76.9%
Minimum	61.8% — Bihar
Maximum	94.0% — Kerala
Range	32.2 percentage points
Missing values / duplicate rows	0 / 0

Table 2: Overall descriptive results calculated from the supplied CSV

Highest and lowest values

The five lowest values are Bihar (61.8%), Arunachal Pradesh (65.4%), Rajasthan (66.1%), Jharkhand (66.4%), and Andhra Pradesh (67.0%). The five highest are Kerala (94.0%), Lakshadweep (91.8%), Mizoram (91.3%), Goa (88.7%), and Tripura (87.2%). These comparisons describe the supplied 2011 figures; they do not establish the reasons for the differences.

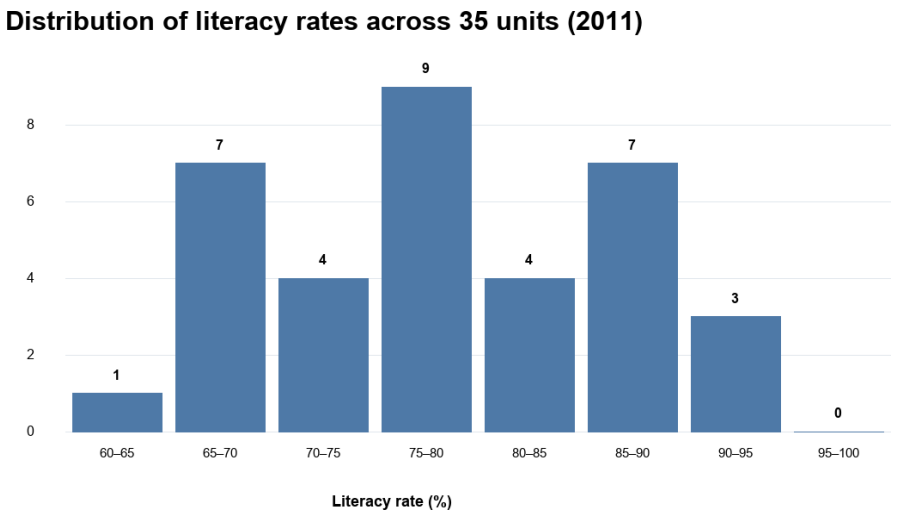


Figure 6: Distribution of the 35 literacy-rate values in the CSV

State and union territory group summary

The 28 state records have a mean of 75.89% and a median of 75.7%. The 7 union territory records have a mean of 85.67% and a median of 86.2%. These group means are descriptive and unweighted. They should not be interpreted as a comparison of equal-sized populations because the groups and their constituent units differ.

Administrative unit	Count	Mean	Median	Minimum	Maximum
State	28	75.89%	75.7%	61.8%	94.0%
Union Territory	7	85.67%	86.2%	76.2%	91.8%

Table 3: Descriptive literacy-rate statistics by administrative-unit type

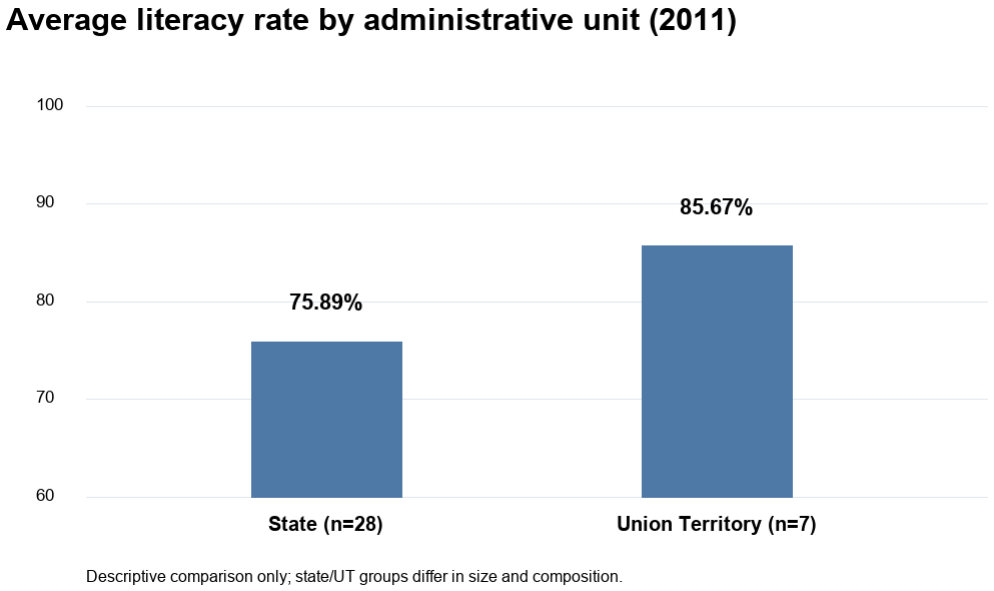


Figure 7: Mean literacy rate by administrative-unit type (generated from CSV)

Observation: The mean for the listed union territories is about 9.78 percentage points higher than the mean for the listed states. This reflects only the values and unit composition in this dataset.

Interpretation

The unit-level data show a broad range rather than a narrow cluster: rates span 32.2 percentage points, and the distribution chart places the largest number of units in the 75–80% band. The high values near 90% and above occur in several units, while the lowest observations are concentrated among a small number of states. A map can help locate these observations, but the ranked chart is more dependable for identifying exact values.

Limitations

The dataset covers only the 2011 census year; it cannot show change over time.

The file contains no population counts, so overall means are unweighted across administrative units.

There are no coordinates or boundary geometries. Geocoded maps may omit or misplace records, as suggested by the two unknown locations visible in the supplied screenshots.

The screenshots show Tableau rather than Power BI, and the map screenshots do not show exact rate labels for every mark.

The labels and administrative units correspond to the supplied Census 2011 dataset; newer administrative arrangements should not be silently substituted.

POWER BI MAP OUTPUTS

1. Purpose and note on the outputs

This annexure adds the geographic outputs that the main report describes conceptually: a bubble (point) map, a filled-style map, and a combined dashboard, all built from the 35 literacy-rate values of the supplied dataset. The visuals use a Power BI-style design (blue colour gradient, bubble size and colour encoding, KPI cards beside a map and ranked bar chart). They were **generated from the dataset values** and are **not screenshots taken inside Power BI Desktop**; the report's own screenshots remain Tableau captures, as already stated in Section 2.4. Section A.5 lists the exact steps to rebuild each visual in Power BI Desktop and capture genuine screenshots.

Because the dataset has no coordinates, the bubble positions use approximate state/UT centroid coordinates (Table 4). The outline of India is a simplified schematic and is not an authoritative boundary. Rates were read from the ranking chart in Figure 5 and reproduce every statistic in Section 6 (mean 77.85%, median 76.9%, state mean 75.89%, UT mean 85.67%).

2. Bubble map (point map)

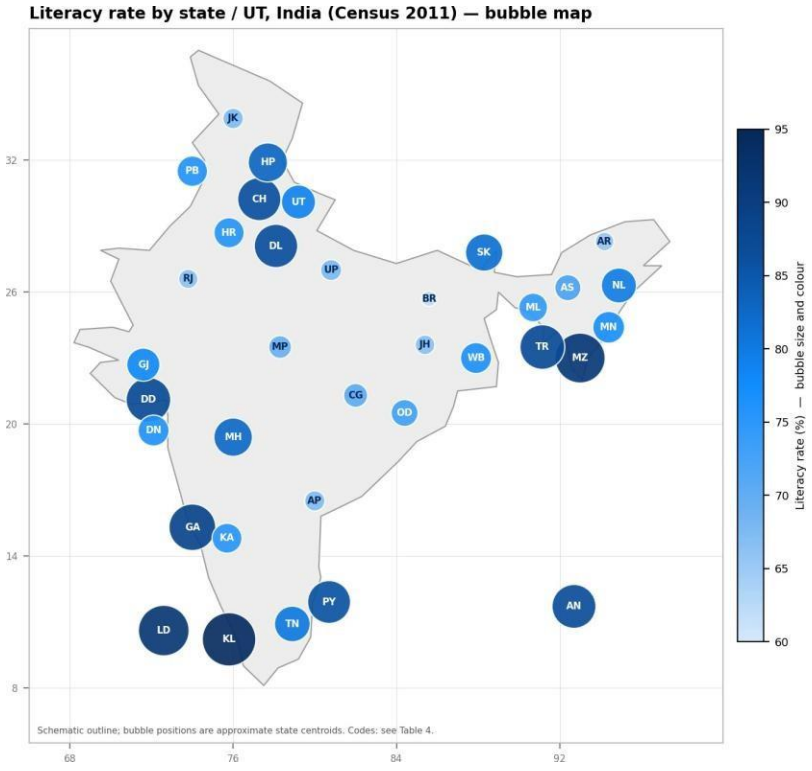


Figure 8: Bubble map of literacy rate by state/UT; size and colour both encode the rate (generated from the CSV values)

Observation: Darker, larger bubbles mark the highest rates: Kerala (94.0%), Lakshadweep (91.8%), Mizoram (91.3%) and Goa (88.7%). The lightest, smallest bubbles mark Bihar (61.8%), Arunachal Pradesh (65.4%), Rajasthan (66.1%) and Jharkhand (66.4%). Low values appear in several northern and central states, while many south-western and island units are high. This is descriptive only and does not explain why the rates differ.

A.3 Filled-style map (tile-grid)

A true filled (choropleth) map needs state/UT boundary polygons matching the 2011 units, which the dataset does not contain. As a substitute, Figure 9 uses a tile-grid map in which each unit is one equal-sized tile arranged in approximate geographic position and filled by literacy rate. Equal tiles prevent large states from dominating the colour impression, and tiny units such as Delhi, Chandigarh and Daman & Diu remain visible. In Power BI the Filled map visual produces the boundary version once the location names are matched.

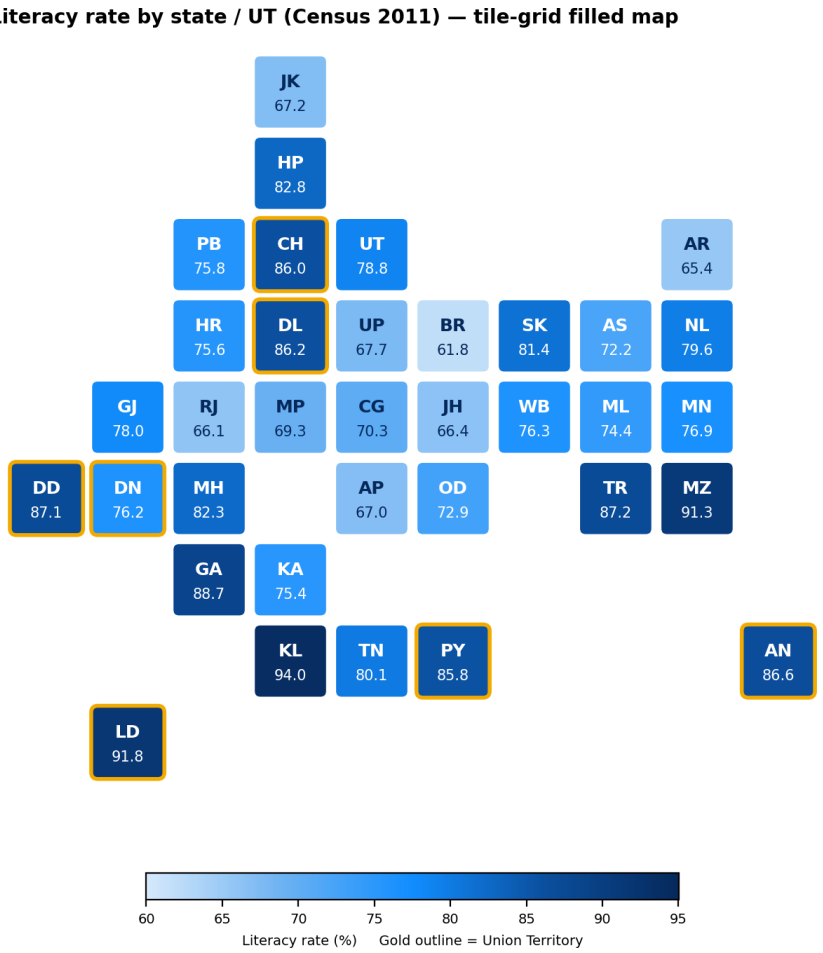


Figure 9: Tile-grid filled map; tile position is approximate and gold outlines mark union territories (generated from the CSV values)

Observation: The filled view makes the union-territory pattern easy to see: all seven UTs are mid-to-dark blue, consistent with their higher mean of 85.67% against 75.89% for states. The lightest tiles, Bihar, Arunachal Pradesh, Rajasthan and Jharkhand, sit in the north and east of the grid. Tile positions are schematic, so distances and borders should not be read from this figure.

Combined dashboard

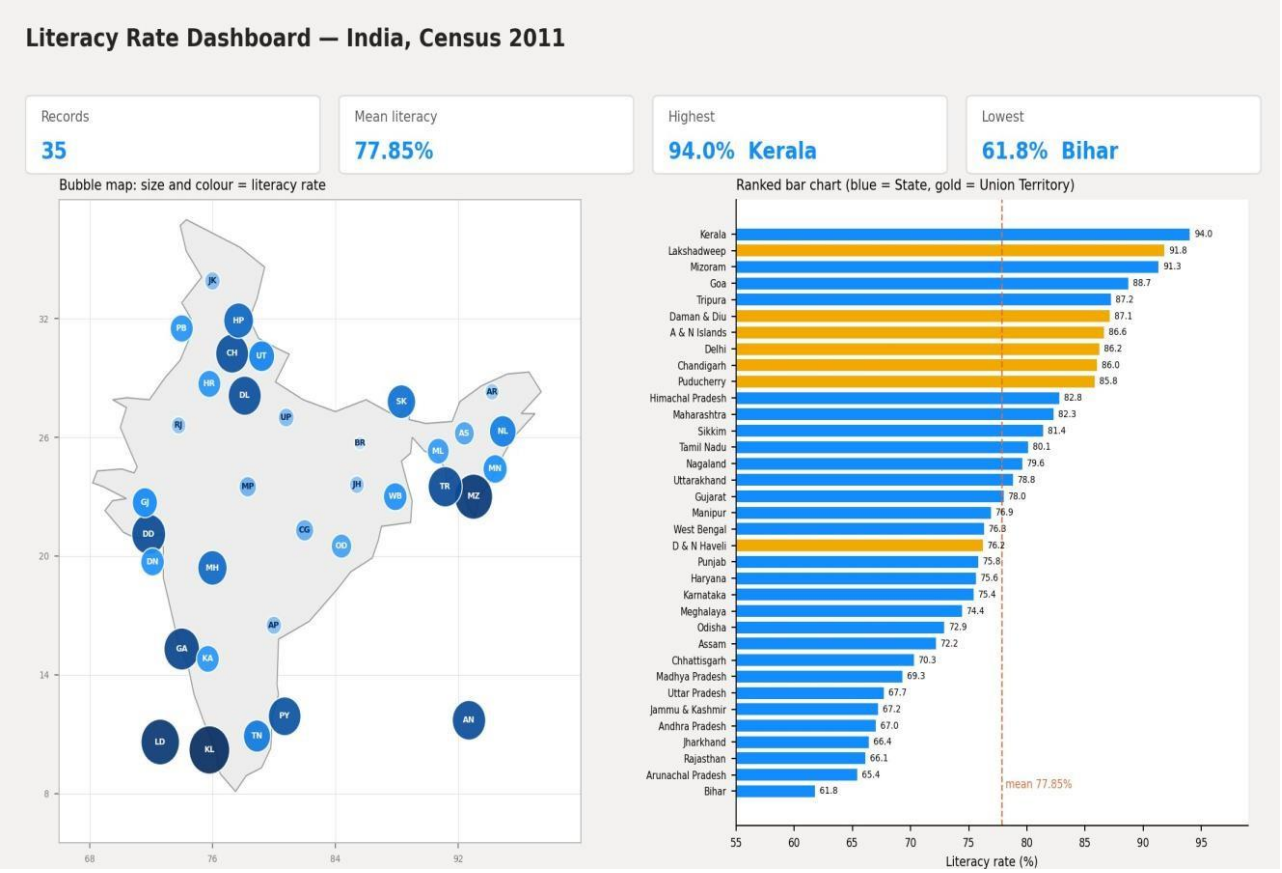


Figure 10: Power BI-style dashboard combining KPI cards, bubble map and ranked bar chart (generated from the CSV values)

Table 4: Location table used for the map outputs

Code	State / UT	Type	Literacy (%)	Latitude	Longitude
AN	A & N Islands	UT	86.6	11.70	92.70
AP	Andhra Pradesh	State	67.0	16.50	80.00
AR	Arunachal Pradesh	State	65.4	28.00	94.20
AS	Assam	State	72.2	26.20	92.90
BR	Bihar	State	61.8	25.70	85.60
CH	Chandigarh	UT	86.0	30.73	76.78
CG	Chhattisgarh	State	70.3	21.30	82.00
DN	D & N Haveli	UT	76.2	20.20	73.00
DD	Daman & Diu	UT	87.1	20.40	72.85
DL	Delhi	UT	86.2	28.60	77.20
GA	Goa	State	88.7	15.30	74.00
GJ	Gujarat	State	78.0	22.70	71.60
HR	Haryana	State	75.6	29.10	76.10
HP	Himachal Pradesh	State	82.8	31.90	77.30
JK	Jammu & Kashmir	State	67.2	33.90	76.00
JH	Jharkhand	State	66.4	23.60	85.40
KA	Karnataka	State	75.4	14.80	75.70
KL	Kerala	State	94.0	10.40	76.40
LD	Lakshadweep	UT	91.8	10.60	72.60
MP	Madhya Pradesh	State	69.3	23.50	78.30
MH	Maharashtra	State	82.3	19.40	76.00
MN	Manipur	State	76.9	24.80	93.90
ML	Meghalaya	State	74.4	25.50	91.30
MZ	Mizoram	State	91.3	23.30	92.80
NL	Nagaland	State	79.6	26.10	94.40
OD	Odisha	State	72.9	20.50	84.40
PY	Puducherry	UT	85.8	11.90	79.80
PB	Punjab	State	75.8	31.00	75.30
RJ	Rajasthan	State	66.1	26.60	73.80
SK	Sikkim	State	81.4	27.60	88.50
TN	Tamil Nadu	State	80.1	11.10	78.50
TR	Tripura	State	87.2	23.80	91.70
UP	Uttar Pradesh	State	67.7	27.00	80.80
UT	Uttarakhand	State	78.8	30.10	79.20
WB	West Bengal	State	76.3	23.00	87.90

Coordinates are approximate centroids chosen for display (small units are nudged apart for legibility). A machine-readable copy is supplied as india_literacy_2011_with_coordinates.csv.

Rebuilding the outputs in Power BI Desktop

1. Load the data. Home > Get data > Text/CSV, choose the dataset (or the coordinates CSV supplied with this annexure) and click Load. Confirm that Literacy_Rate_Percent is a decimal number.

2. Enable and categorise geography. Enable File > Options > Security > Use Map and Filled Map visuals. In Data view select State_or_UT and set Column tools > Data category to *State or Province*; set Country to *Country/Region*. If Latitude and Longitude columns are loaded, mark them as *Latitude* and *Longitude*.

3. Bubble map. Insert the Map visual. Put State_or_UT in Location (and Country in the location hierarchy above it), Literacy_Rate_Percent in Bubble size, and Administrative_Unit in Legend or apply a gradient under Format > Bubbles > Colors > fx (minimum 61.8, maximum 94.0). Add State_or_UT and the rate to Tooltips.

4. Filled map. Insert the Filled map visual with State_or_UT in Location and Literacy_Rate_Percent in Color saturation. Set the gradient under Format > Fill colors > fx. The Azure Maps filled-map visual (Reference [4]) is an alternative with boundary options.

5. Ranked bar chart and cards. Insert a clustered bar chart with State_or_UT on the axis and Literacy_Rate_Percent as the value; sort descending and turn on data labels. Add four Card visuals: Count of State_or_UT, Average, Max and Min of Literacy_Rate_Percent. Arrange the cards, map and bar chart as in Figure 10; clicking a bar cross-highlights the map.

6. Resolve unmatched locations. The Tableau screenshots reported two unknown locations. The most likely causes are the abbreviated names “A & N Islands”, “D & N Haveli” and “Daman & Diu”; this is a hypothesis to verify, not a finding. Replace them with “Andaman and Nicobar Islands”, “Dadra and Nagar Haveli” and “Daman and Diu” (or use the latitude/longitude columns, which avoid geocoding). Confirm that all 35 locations plot before treating the map as complete.

7. Capture the outputs. Use File > Export > Export to PDF, or the Snipping Tool (Windows + Shift + S), to capture each page and replace Figures 8 to 10 with the real Power BI screenshots if required by the course.

CONCLUSION

This micro project examined how geographic visualization can support comparison of literacy rates across Indian states and union territories. The supplied CSV contains 35 complete records, and the analysis calculated an overall mean of 77.85%, a median of 76.9%, and a range from 61.8% in Bihar to 94.0% in Kerala. A location map gives spatial context, while a sorted bar chart makes the values easier to compare.

The supplied screenshots show a Tableau data-source view, bar chart, point map, and dashboard. They demonstrate the visualization arrangement available in the provided evidence, but they do not document execution in Power BI. Power BI's map concepts can support a similar report design, provided the state/UT geographic categorization and unmatched locations are validated. A map should not be treated as complete while two records remain unknown.

The analysis is limited to one census year and does not include population weighting or variables that could explain geographic differences. Future work can add validated 2011 boundary data, inspect and resolve the two unmatched names, include population counts for weighted summaries, and compare multiple census years when consistent data are available. These additions would improve geographic accuracy and broaden the analysis while retaining the transparent comparisons made here.

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